

RAKSHA-NETRA

EDGE-AI × MULTIMODAL SENSING × CYBERSECURITY

Rapid presumptive trace screening for railway security environments

EVIDENCE STATUS This repository presents a proposed engineering architecture and virtual prototype package. It does not claim validated field performance or definitive chemical identification. The validation roadmap defines the evidence required before operational deployment.

COLLECT Controlled sample	ANALYZE Local sensing + Edge AI	ALERT Presumptive state	SECURE Protected event	REPORT Dashboard / sync
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01 / PROJECT AT A GLANCE

RAKSHA-NETRA is a proposed rugged, handheld, Edge-AI-enabled presumptive trace-screening platform. It combines VOC, electrochemical and optical sensing with local sensor fusion, on-device inference, secure event recording and dashboard-oriented reporting.

Design principle: Keep the critical screening workflow close to the operator while preserving a secure and traceable digital record.

01 PORTABLE Rugged handheld field concept	02 EDGE AI Local multimodal inference	03 OFFLINE-FIRST Connectivity-gap resilience	04 SECURE LOG Cryptographically verifiable history
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SYSTEM THESIS

SAMPLE Controlled collection	SENSORS VOC • EC • Optical	STM32 Acquisition + control	RPI 5 Preprocess • fuse	TFLITE Local inference	SECURE LOG AES-256 • SHA-256	SYNC Protected queue → dashboard
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Continuous evidence chain: controlled collection feeds multimodal sensing; deterministic device control is separated from higher-level fusion and inference; secure logs and offline storage preserve continuity until synchronization is available.

02 / SYSTEM ARCHITECTURE & ENGINEERING INTENT

The architecture separates time-critical sensing and control from higher-level compute and communication. This makes the core workflow easier to reason about while leaving the AI and sensing layers extensible.

LAYER	RESPONSIBILITY	DESIGN INTENT
Sampling / handling	Controlled sample collection	Keep operator workflow repeatable and traceable.
STM32	Real-time acquisition, heating/control, safety and diagnostics	Deterministic low-level control.
Raspberry Pi 5	Preprocessing, multimodal fusion, storage and communications	Higher-level compute and orchestration.
Edge AI	TensorFlow Lite local inference	Reduce dependence on continuous cloud connectivity.
Secure event log	Encryption, hashing, sequence linkage, metadata	Protect and contextualize recorded screening events.
Dashboard	Reporting and monitoring	Surface synchronized screening records for downstream review.
Modularity	Modular sensing layer	Pathway for future sensor cartridges and dataset/model expansion.
Resilience	Offline-first storage	Temporary network loss does not stop the core workflow.

ENGINEERING SPLIT

SAMPLE Controlled collection	SENSORS VOC • EC • Optical	STM32 Acquisition • control	RPI 5 Fusion • preprocessing	TFLITE Local inference	SECURE LOG Protected record
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03 / MULTIMODAL SENSING STACK

RAKSHA-NETRA uses multiple sensing modalities so that the screening decision is based on a multimodal response pattern rather than a single sensing channel. The intended output is a presumptive classification/alert state, not definitive identification.

SENSING CHANNEL	CORE COMPONENTS	ROLE IN THE PLATFORM
VOC response	BME688 + SGP40	Capture volatile-response patterns

Electrochemical	SPCE + AD5940	Measure electrochemical responses
Optical / colorimetric	TCS34725 + Camera	Capture optical/colorimetric changes

MULTIMODAL FINGERPRINT

The sensing layer combines complementary response information across VOC, electrochemical and optical channels, creating a pathway for fused feature representations and future cartridge/model expansion.

- Different sensing channels can provide complementary response information.
- A fused representation can use patterns across VOC, electrochemical and optical responses.
- The modular structure allows future cartridges, model revisions and dataset expansion without redefining the whole system.

PRACTICAL BOUNDARY

Sensor responses may be affected by environmental conditions, contamination, calibration drift and sample variability. Independent validation and authorized confirmatory procedures are required before operational use.

04 / EDGE-AI DECISION PIPELINE

The decision path is designed to remain local. Synchronized readings from the sensing modalities are conditioned, transformed into features or embeddings, fused, and passed to a TensorFlow Lite model running on Raspberry Pi 5.

01 ACQUIRE Synchronized streams	02 CONDITION Preprocess	03 EXTRACT Features / embeddings	04 FUSE Multimodal representation	05 INFER TensorFlow Lite	06 PRESENT Presumptive state	07 RECORD Secure evidence
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STEP	STAGE	DESCRIPTION
01	Acquire	Acquire synchronized readings from multiple sensing modalities.
02	Condition	Condition and preprocess individual sensor streams.
03	Extract	Extract features or embeddings from each modality.
04	Fuse	Fuse the modalities into a multimodal representation.
05	Infer	Run TensorFlow Lite inference locally on Raspberry Pi 5.
06	Present	Present a presumptive classification / alert state to the operator.
07	Record	Record the event and supporting metadata through the secure logging layer.

EDGE-FIRST OUTCOME

The AI decision is designed to remain available during connectivity gaps; network services become a synchronization and reporting path rather than a hard dependency for the core screening flow.

05 / SECURITY BY DESIGN

Security is treated as part of the evidence pipeline, not as an afterthought. Each screening event is contextualized with provenance metadata and protected by cryptographic controls.

EVENT Device ID • time • GPS	PROTECT AES-256	INTEGRITY SHA-256	LINK Hash chain	RESULT Traceable history
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CONTROL	IMPLEMENTATION	VALUE
Confidentiality	AES-256	Protect stored / transmitted sensitive data
Integrity	SHA-256	Detect content modification
Record linkage	Hash-chained events	Make unauthorized sequence changes detectable
Provenance	Device ID + timestamp + GPS	Contextualize each screening event
Availability	Offline-first local storage	Continue operation through connectivity gaps

IMPORTANT DISTINCTION

Hash-chaining provides tamper-evidence and record linkage; it does not by itself establish truthfulness of the underlying sensor measurement. Measurement validity still depends on calibration, environmental controls and independent validation.

06 / OFFLINE-FIRST OPERATION & DASHBOARD

When connectivity is unavailable, screening events can remain on-device in protected local storage. When connectivity returns, queued information can be synchronized according to the defined backend policy.

DEVICE Screening workflow	EVENT LOG Protected local record	CONNECTIVITY Wi-Fi / 4G-LTE	SYNC Queued transfer	RPF DASHBOARD Report + monitor
OPERATIONAL STATE		EXPECTED BEHAVIOUR		
Connected		Screening events can proceed and be synchronized according to backend policy.		
Temporarily offline		Core screening workflow continues; events remain in protected local storage.		
Connectivity restored		Queued information can be synchronized.		
Operator review		Dashboard provides a reporting/monitoring surface for synchronized events.		

Design intent: Offline-first is not simply a connectivity feature; it is a continuity strategy so a temporary network gap does not interrupt the core field workflow.

07 / PROTOTYPE EXPERIENCE & DIGITAL EVIDENCE

The Figma prototype demonstrates the intended operator journey from device readiness through controlled sample handling, multimodal analysis, presumptive threat state, secure event

STEP	STAGE	WHAT THE PROTOTYPE COMMUNICATES
01	DEVICE READY	Operator begins from a ready state.
02	CONTROLLED SAMPLE	Sample-handling stage shown in the operator flow.
03	MULTIMODAL ANALYSIS	Multiple sensing channels feed the local processing path.
04	PRESUMPTIVE STATE	Operator receives a screening / alert state.
05	SECURE EVENT LOG	Screening event is recorded with supporting metadata.
06	RPF DASHBOARD	Synchronized information is surfaced for reporting and monitoring.

08 / TECHNOLOGY MATRIX

LAYER	TECHNOLOGY / ASSET	PURPOSE
MCU	STM32	Deterministic acquisition, control, safety and diagnostics
Edge computer	Raspberry Pi 5	Preprocessing, multimodal fusion, storage and communications
AI inference	TensorFlow Lite	On-device model execution
VOC sensing	BME688 • SGP40	Volatile-response measurements
Electrochemical	SPCE • AD5940	Electrochemical response measurements
Optical	TCS34725 • Camera	Optical / colorimetric capture
Connectivity	Wi-Fi • 4G/LTE	Data transfer and synchronization
Protocol layer	MQTT / REST	Device-to-backend communication
Security	AES-256 • SHA-256 • Hash-chained logs	Confidentiality, integrity, sequence linkage
Design / prototype	EasyEDA • Figma	Circuit representation and operator-flow prototype

DESIGN ATTRIBUTES

PORTABLE Rugged handheld field concept	MODULAR Sensing + model expansion	EDGE Local fusion + inference	SECURE Protected linked history	RESILIENT Offline-first operation
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09 / INNOVATION SNAPSHOT

DESIGN ELEMENT	WHAT IT CONTRIBUTES
Multimodal chemical-response fingerprinting	Uses VOC, electrochemical and optical response information rather than relying on one sensing channel.
Edge-AI inference	Keeps local decision support closer to the operator and reduces dependence on continuous cloud connectivity.
Hardware / software separation	Deterministic control is separated from flexible AI processing, enabling iterative model evolution.
Cryptographically verifiable history	Adds confidentiality, integrity checks, sequence linkage and provenance metadata to screening records.
Modular sensing layer	Creates a pathway for future sensor cartridges, datasets and models.
Device-to-dashboard workflow	Extends the concept from a sensor device to an end-to-end operational information flow.

SYSTEM-LEVEL INNOVATION

The core concept is a continuous evidence chain: collect → analyze → alert → securely record → report. Each stage is tied to the next, so the technical design reads as a single system rather than a collection of disconnected technologies.

10 / VALIDATION ROADMAP

The current brief defines the architecture and prototype experience; the next step is to convert the design into measurable engineering evidence. The roadmap below makes the evidence boundary explicit.

PHASE	FOCUS	EVIDENCE TARGET
01 • Prototype	Integrate sensing, MCU, edge computer and UI	Working proof-of-concept
02 • Dataset	Controlled multimodal response collection	Labeled dataset
03 • Model	Training, validation, quantization	Edge-compatible model
04 • Security	Threat modeling and integrity testing	Hardened event pipeline
05 • Field validation	Environmental and operational testing	Performance evidence

11 / GITHUB REPOSITORY MAP

A national-level hackathon repository should let a reviewer understand the project quickly, then inspect deeper technical evidence without hunting for files. The recommended structure keeps the README concise while making engineering assets discoverable.

PATH	PURPOSE
Documentation/	Project brief, architecture, specifications and supporting reports
Hardware/	Schematic, PCB and bill of materials
Software/	STM32 and Raspberry Pi implementation
AI_Model/	Preprocessing, model assets and inference notes
Dashboard/	Dashboard / backend prototype and integration notes
Prototype/	Figma exports, screenshots and interaction flows
Simulation/	Wokwi or other simulation artifacts, when applicable
Presentation/	Hackathon presentation and supporting media

12 / RESPONSIBLE USE, LIMITATIONS & DEPLOYMENT BOUNDARY

RAKSHA-NETRA is a prototype for presumptive screening and decision support. Sensor responses can be affected by environmental conditions, contamination, calibration drift and sample variability. A presumptive result must not be treated as definitive identification.

BOUNDARY	REQUIRED PRACTICE
Presumptive result	Treat the result as a screening state, not definitive identification.
Calibration	Use appropriate calibration and monitor drift over time.
Environmental effects	Test across relevant temperature, humidity, contamination and sample-variability conditions.
Confirmatory action	Use appropriate confirmatory testing and authorized procedures before operational deployment.
Model governance	Maintain traceable model versions and document validation evidence.
Deployment	Independent validation is required before operational deployment.

RAKSHA-NETRA

A secure, portable and intelligent approach to presumptive field screening.

EDGE AI · MULTIMODAL SENSING · OFFLINE-FIRST · CRYPTOGRAPHIC TRACEABILITY